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LECTURE TEXT

Square-wave voltammetry

Valentin Mirceski[1,2] · Sławomira Skrzypek[1] · Leon Stojanov[2]

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Abstract
This text is written as a very basic, first introduction to square-wave voltammetry, as one of the very specific, but most versatile techniques in the family of pulse voltammetric techniques. To make the text self-consistent, a brief introduction to voltammetry is initially given, to make the next elaboration of square-wave voltammetry more easily understandable. For more comprehensive introduction to voltammetric techniques, the previous chem-text is recommended (Scholz in ChemTexts 1:17, 2015). The aim of the current text is to explain the basic features of square-wave voltammetry, to reveal the essential idea behind the specific shape of the excitation signal used, as well as to demonstrate that square-wave voltammetry is highly suited to study the mechanism of electrode reactions. The latter is illustrated by addressing the simplest experimental situation encountered in voltammetry to make the text didactically suited for broader student population studying chemistry and related sciences.

Keywords Square-wave voltammetry · Pulse voltammetry · Mechanism of electrode reactions · Chemistry

Voltammetry: a brief overview

Voltmammetry is based on microelectrolysis

Voltammetry encompasses a large group of electrochemical techniques where nonspontaneous, interfacial *charge transfer* processes (electron or ion transfer across the electrode|electrolyte solution interface) are driven by externally applied electrical potential difference in an electrolytic cell [1, 2]. The aim of voltammetry frequently is characterisation of redox species and the analytical determination. Additionally, the thermodynamics and kinetics of the charge transfer can be studied and possibly coupled chemical reactions or adsorption phenomena. Voltammetric experiments are frequently conducted in a few millilitres of an electrolyte solution, using small electrodes with a surface area less than a square centimetre; alternatively, in more specific cases, electrodes are used with dimensions ranging from the micrometre to nanometre scale [3]. As redox reactions of the studied electroactive species take place at the electrode|electrolyte interface only, minute amounts of the electroactive reactant are consumed and the overall chemical composition of the system remains practically constant.

The voltammetric measurement is organised in a specific way to provide information on the phenomena taking place at one of the electrodes only, which is termed *working electrode*. A scheme of a typical cell is depicted in Fig. 1, consisting of the working electrode (WE), with the smallest electrode surface area, the second electrode, called *reference electrode* (RE), and the third, the *counter electrode* (CE). A current is flowing only between the working and the counter electrodes. The working electrode is the bottleneck for the current flow; thus it carries information solely from the charge transfer across the small working electrode|electrolyte interface. Electrochemical reactions at the counter electrode are usually unknown, and finally irrelevant for the outcome of the measurement.

The reference electrode should be robust, with a constant chemical composition, which determines its constant electrical potential, [4]. Its role is to serve as a reference point to control the relative potential of the working electrode with the aid of an instrument called *potentiostat*. The *electrical potential* difference between the solution and the working

✉ Valentin Mirceski
valentin@pmf.ukim.mk

[1] Department of Inorganic and Analytical Chemistry, Faculty of Chemistry, University of Lodz, Tamka 12, 90-403 Lodz, Poland

[2] Institute of Chemistry, Faculty of Natural Sciences and Mathematics, P.O. Box 162, 1000 Skopje, Republic of Macedonia

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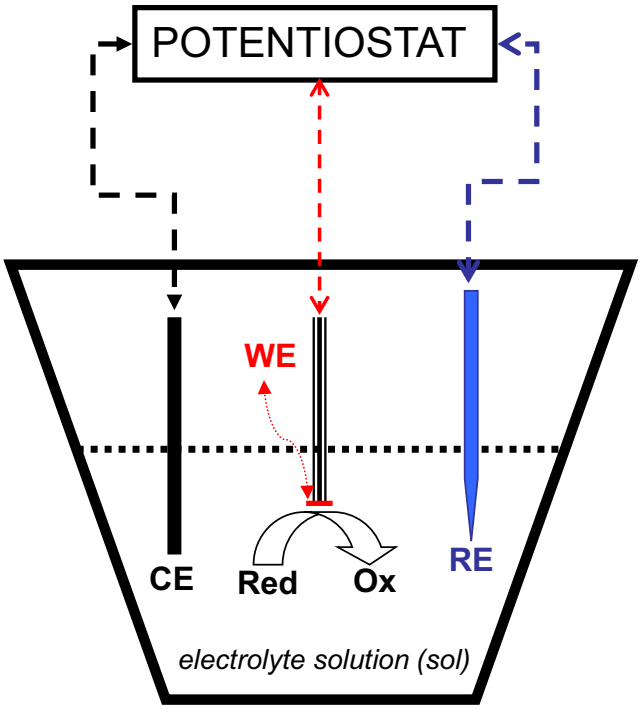


Fig. 1 Scheme of a voltammetric cell where the interfacial charge transfer is a redox reaction at the working electrodelelectrolyte interface, and the current flows between the small *working electrode* (WE) and the large *counter electrode* (CE). The potential of the working electrode is controlled versus the constant potential of the robust *reference electrode* (RE) with the aid of the *potentiostat*

electrode is the driving force for the electrochemical reaction and the resulting flow.

> Most potentiostats for 3-electrode measurements are working this way that the potential of the electrolyte solution is equal to the potential of the metallic phase of the reference electrode. The potential difference between the metallic phase of the working electrode and the electrolyte solution is thus equal to that between the working and the reference electrode. The voltage between the working and the counter electrode is adjusted by the potentiostat so as to have the desired potential difference between the working electrode and electrolyte solution (= reference electrode). For voltammetric experiments, i.e., where the potential is a certain function of time, the potentiostat has to be interfaced with a potential wave form generator.

The foregoing discussion implicitly follows that in voltammetry one plays with the physical parameters such as *electrical potential* and *current*. The former is an independent variable, controlled by the instrumentation, whereas the latter is the dependent one (i.e., an outcome of the experiment). Hence, the term "voltammetry" is coined from abbreviations referring to the units, i.e., volt (V) and ampere (A), plus the suffix "metry" with a meaning of measuring; hence, $volt + am + metry \equiv voltammetry$.

The simple shape of the voltammogram reflects a variety of interfacial phenomena

A graphical representation of the outcome of a voltammetric experiment is called *voltammogram* (Fig. 2a). It is an *I–E* curve, representing the variation of the current *I* versus the controlled potential *E* of the working electrode, relative to the constant potential of the reference electrode. In Fig. 2a, the simplest form of a voltammetric experiment has been assumed. The potential is varied linearly in time within a certain potential interval around the formal potential ($E^{\emptyset\prime}$) (Fig. 2b) [3]. This sort of voltammetry is known as voltammetry with a linear potential scan or *linear sweep voltammetry* (LSV). The main parameter of the technique is the rate of the potential variation with time (sweep rate, or scan rate, *v*). The sweep rate is defined as a ratio of the potential interval ΔE ($\Delta E = E_f - E_s$, where E_f and E_s is the final and starting potential, respectively) and the time interval Δt (i.e., the time required to cross the potential path ΔE), thus $v = \Delta E/\Delta t$. Obviously, *v* determines the overall time of the experiment, which could range from minutes to milliseconds.

In Fig. 2, it is further assumed that the experiment is conducted at a static working electrode (e.g., chemically inert platinum electrode with a surface area in the range of square centimetres, i.e., *macroscopic electrode*). The electrode is immersed in an unstirred solution. Most frequently, water, or some other polar organic liquid is the solvent. The solvent contains dissolved electroactive compound [e.g., hexacyanoferrate(II) ions] typically in millimolar concentration and a supporting electrolyte (e.g., KNO_3), commonly at a concentration at least one order of magnitude higher. The theoretically predicted voltammogram has a simple, asymmetric peak shape (Fig. 2a).

Although the voltammetric curves may look very simple, a variety of physicochemical processes (e.g., diffusion, migration, convection) and chemical reactions (redox reactions and accompanying chemical reactions) can be involved. This can make a correct interpretation of the data quite challenging. The simplest possible voltammetric experiment envisaged in Fig. 2 is at least affected by intrinsic thermodynamic features of the interfacial *redox reaction* (1) (e.g., Gibbs free energy) and the mass transfer phenomenon known as diffusion [3]:

$$\mathrm{Red}_{sol} \rightleftarrows \mathrm{Ox}_{sol} + ne^{-}. \quad (1)$$

For the sake of simplicity, the charges of the species are omitted and the abbreviation (sol) indicates dissolved species, whereas *n* is the stoichiometric number of electrons. The redox reaction (1) takes place at the working electrodelelectrolyte interface only. It is *heterogeneous*

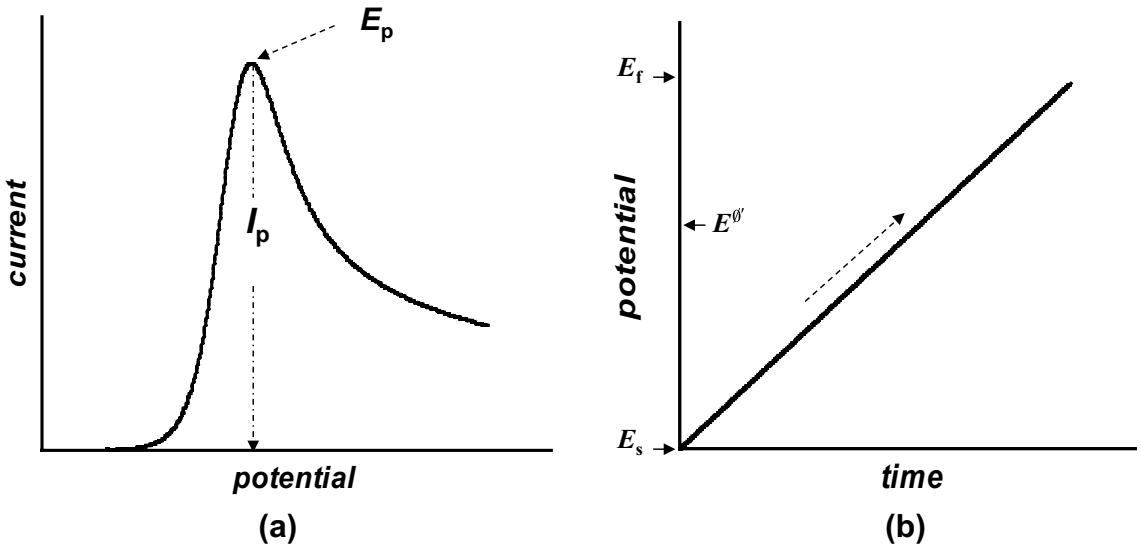


Fig. 2 **a** Typical voltammogram obtained at a macroscopic working electrode by varying the electrode potential in the simplest linear fashion from certain starting (E_s) to the final (E_f) potential. **b** The time variation of the potential at a certain rate v, known as the *scan rate*. Such technique is called *linear sweep voltammetry*. The maximum of the asymmetric peak-shaped voltammogram, cf. **a**, is defined by the *peak-current* (I_p) and the *peak-potential* (E_p)

interfacial chemical process, as the two participants Ox_{sol} and Red_{sol} reside in the electrolyte solution. The third participant, electrons, resides in a different phase, i.e., the working electrode. Electrons are exchanged between the electrode and the redox species across the interface [3]. Thus, the redox reaction (1) is referred to *electrode reaction*, being different to common homogeneous redox reactions between two redox couples in a single phase (e.g., $Red_{1sol} + Ox_{2sol} \rightleftarrows Ox_{1sol} + Red_{2sol}$) [3].

In this context, we can recall again the physical meaning of the electrical potential of the working electrode, which can be alternatively understood as a measure of the *activity of electrons* in the electrode (1). Therefore, one can intuitively assume that variation of the potential enables shifting of the redox equilibrium (1) in a desirable direction, i.e., to the right-hand side (oxidation), or to the opposite side, i.e., the reduction.

At the beginning of the experiment, it is assumed that only Red_{sol} species are homogeneously distributed in the electrolyte solution at a *bulk concentration* designated as $c_{b,Red}$.[1] Accordingly, Ox is absent at the beginning of the experiment ($c_{b,Ox} = 0$). In the course of the experiment, the potential of the working electrode is progressively increased commencing from a lower toward higher values that is $E^{\emptyset'}$. Consequently, the electrode reaction (1) is progressively shifted toward the right-hand side, causing the current to flow. Both Red and Ox species establish certain concentrations at the interface, which differ from the corresponding bulk concentrations. The concentrations at the interface are termed as *surface concentrations*, $c_{s,Red}$ and $c_{s,Ox}$, respectively. As Red_{sol} is consumed and Ox_{sol} is produced, $c_{s,Red} < c_{b,Red}$, $c_{s,Ox} > c_{b,Ox}$, respectively. Hence, concentrations of redox species vary with time and distance. In mathematical terms, they are a function of the two variables, distance (x), measured from the electrode surface, and time (t), measured from the beginning of the experiment, [written as $c(x,t)$]. Hence, a surface concentration can be also noted as $c(0, t)$.

Concentration gradients established in the vicinity of the electrode surface prompt the natural mass transfer known as *diffusion* [3]. The diffusion of the species frequently affects the voltammetric experiment and the shape of the voltammetric curve to a large extend. Due to diffusion, the maximum of the voltammetric curve, known as *peak-current* (I_p), is a linear function of the bulk concentration of the initial electroactive reactant:

$$I_p = k \times c_{b,Red}, \tag{2}$$

where k is the constant for a given experimental conditions. The *linearity* between the peak-current and the bulk concentration provides a basis for *analytical application of voltammetry* [1].

Beyond the maximum, the decaying tail of the curve is also a consequence of diffusion, causing the overall voltammogram to gain an asymmetric peak-like shape. Besides the peak-current, the curve is additionally attributed with another important parameter known as *peak-potential* (E_p).

[1] For simplicity reasons concentration instead of activity.

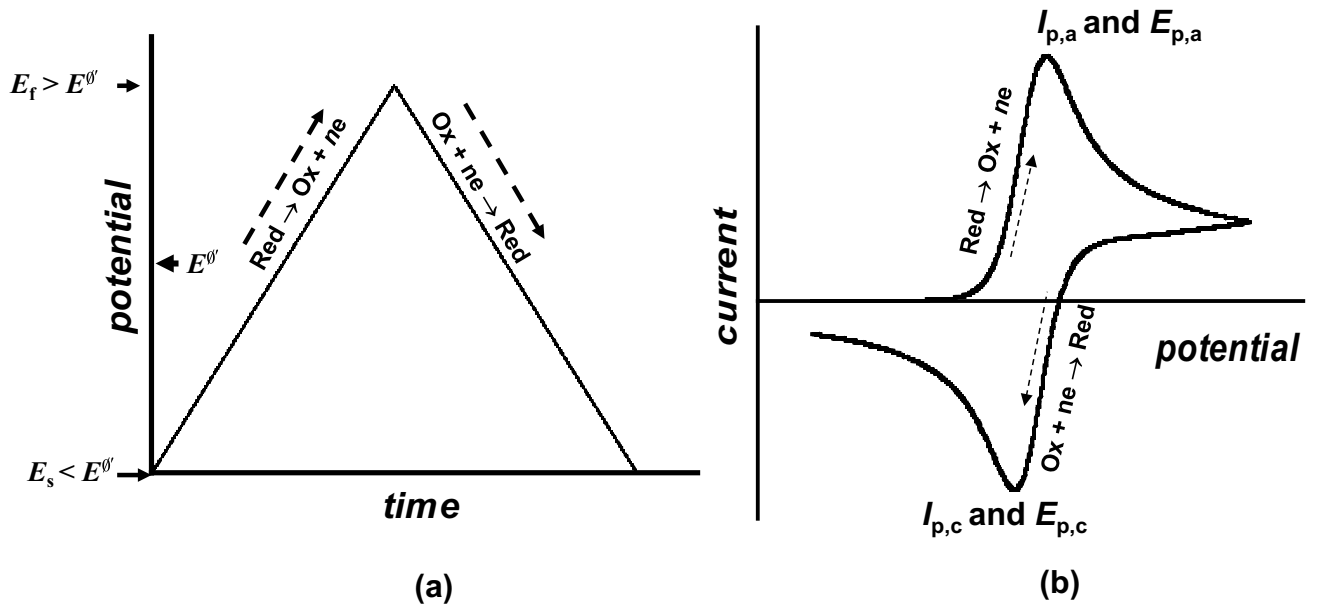


Fig. 3 **a** Typical potential variation and **b** the shape of the *I–E* curve in the popular technique known as *cyclic voltammetry*

It is the position of the maximum at the potential axis. The latter is related to the formal potential ($E^{\emptyset'}$), which is thermodynamic parameter of a critical importance for the electrode reaction (1) [3].

At the end of this section, it has to be mentioned that apart from the foregoing discussed, the outcome of the experiment is frequently additionally affected by the kinetics of the electrode reaction, a variety of surface phenomena, chemical reactions coupled to the electrode reactions, unavoidable electrical phenomena related to the formation of electric double layer at the electrode|electrolyte interface, etc.

The overwhelming beauty of a cyclic voltammogram

At the end of the experiment described in Fig. 2, the electrode reaction (1) is strongly drifted toward the right-hand side, resulting in formation of Ox_{sol} species in the vicinity of the electrode. The initial reactant, Red_{sol}, is exhausted from the electrode region, while the diffusion is trying to resupply it from the bulk. Thus, the interfacial region where the diffusion mass transfer exchanges the material with the bulk is termed as a *diffusion layer*. The decaying tail of the voltammogram (cf. Fig. 2a) reflects the mass transfer in the expanding diffusion layer. In proportion to the time of the experiment, the region becomes richer with Ox_{sol} and poorer with Red_{sol}.

Considering the foregoing facts, one can intuitively predict that the electrode reaction (1) can be drifted in opposite, left-hand side (reduction), if the experiment continues with oppositely oriented potential sweep (Fig. 3a). Such a sort of voltammetric experiment is known as *cyclic voltammetry* (CV), with a typical voltammogram shown in Fig. 3b. Obviously, the first half of the cyclic voltammetry is identical as in Fig. 2, while in the second half, the reaction is drifted in the opposite direction (reduction of Ox_{sol} to Red_{sol}) with the aid of the reverse potential sweep (cf. Fig. 3a). The corresponding *cyclic voltammogram* (Fig. 3b) is attributed with forward and reverse components, reflecting the oxidative and reductive shifts of the electrode equilibrium, respectively. By convention, the oxidation (termed as *anodic*) and reduction (termed as *cathodic*) currents are assigned with a positive and negative sign, respectively.

The main advantage of cyclic voltammetry is its inherent ability to drive the electrode system (1) in both anodic and cathodic directions. Any additional physicochemical processes involving redox species (e.g., adsorption, homogeneous chemical reactions, disproportionation, etc.), could be inspected with cyclic voltammetry. Thus, CV is the main tool in hand to study *electrode mechanisms*, making the technique by far the most popular among electrochemists and chemistry-related community.

Cyclic voltammogram (cf. Fig. 3b) bares overwhelming aesthetic features, reflecting the chemical drift of a given equilibrium in a perfectly precise and controllable way. The latter could be hardly achieved by any other technique in chemistry. Again, the main instrumental parameter is the scan rate. The key features of the voltammogram are the peak-current and peak-potential of both forward (anodic; $I_{p,a}$ and $E_{p,a}$) and backward (cathodic; $I_{p,c}$ and $E_{p,c}$) components (cf. Fig. 3b). By careful inspection of the interrelation of these parameters as a function of the scan rate, one can deduce information regarding thermodynamics and kinetics

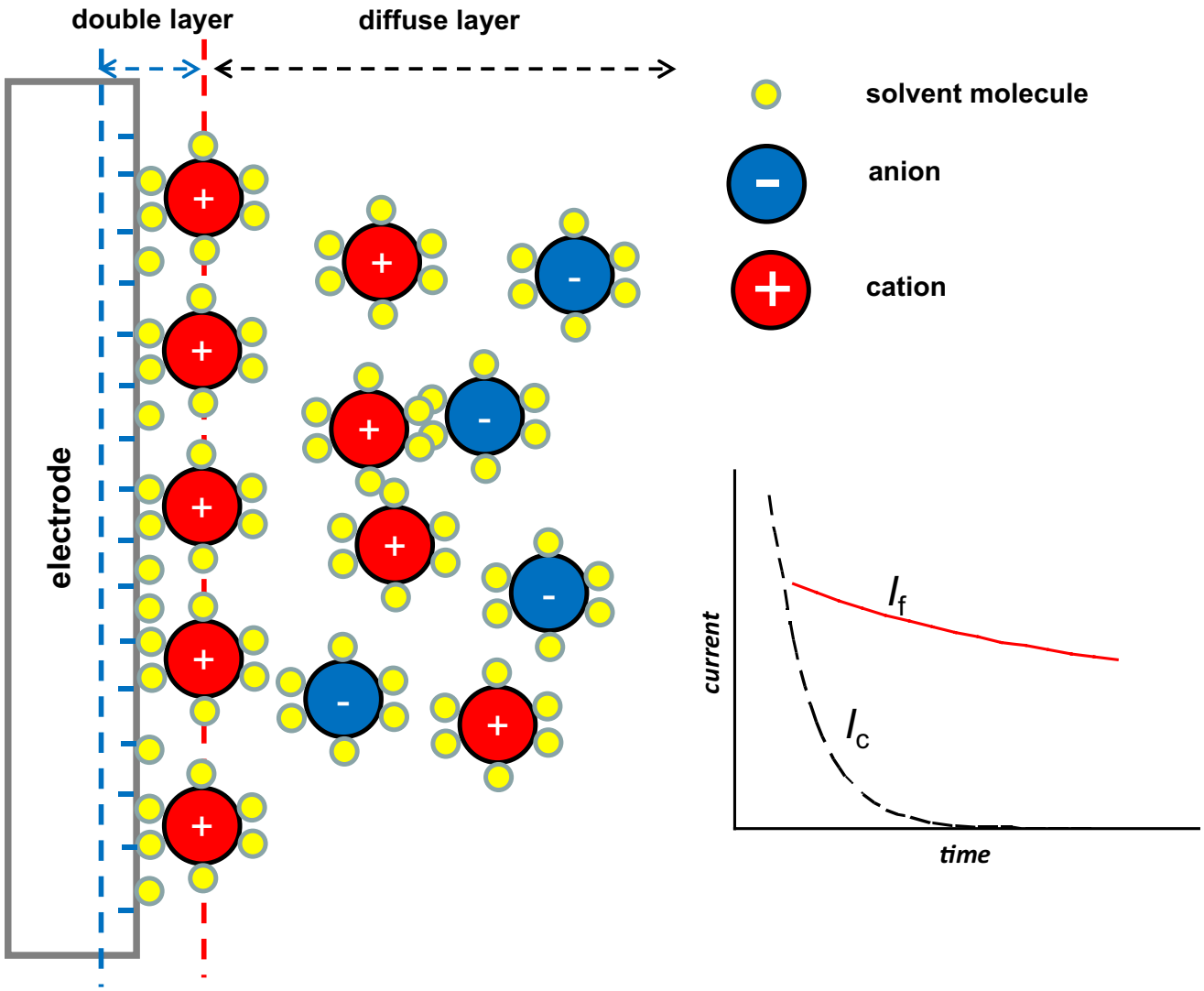


Fig. 4 A primitive sketch of the electric double layer at the electrode|electrolyte interface and the assumed variation of the charging (I_c) and faradaic (I_f) current in time (inset)

of the electrode reaction, the mechanism of interfacial electron transfer, adsorption of redox species, the role of coupled chemical reactions, etc.

A current flows even in the absence of an electrode reaction

Unfortunately, the electrode processes involving charge transfer [e.g., reaction (1)], termed as *faradaic* processes, are not the sole source of current in voltammetry. There are interfacial phenomena which do not involve an interfacial charge transfer, yet giving rise to the current. Such phenomena are consequently termed as *non-faradaic processes*. They are primarily related to the formation of an electric double layer at the electrode|electrolyte interface, which is inevitably present at any given electrode potential. A microscopic double layer is formed due to an excess of charge at the electrode surface and the corresponding response (arrangement) of the counter, free moving charged particles at the side of the electrolyte solution (i.e., electrolyte ions). In addition, polar solvent molecules are also accordingly orientated at the interface (Fig. 4). The formed microscopic double layer acts as a capacitor with a significant capacitance due to extremely small, microscopic distance between the two oppositely charged layers. Thus, formation of the double layer (i.e., charging of the interfacial capacitor) is associated with the current flow. Such current is termed as *charging current* (I_c). The charging current interferes with the faradaic one, because both are the additive constituents of the overall current. The charging current can be even significantly higher than the faradaic one, when the concentration of the electroactive reactant is very low (e.g.,

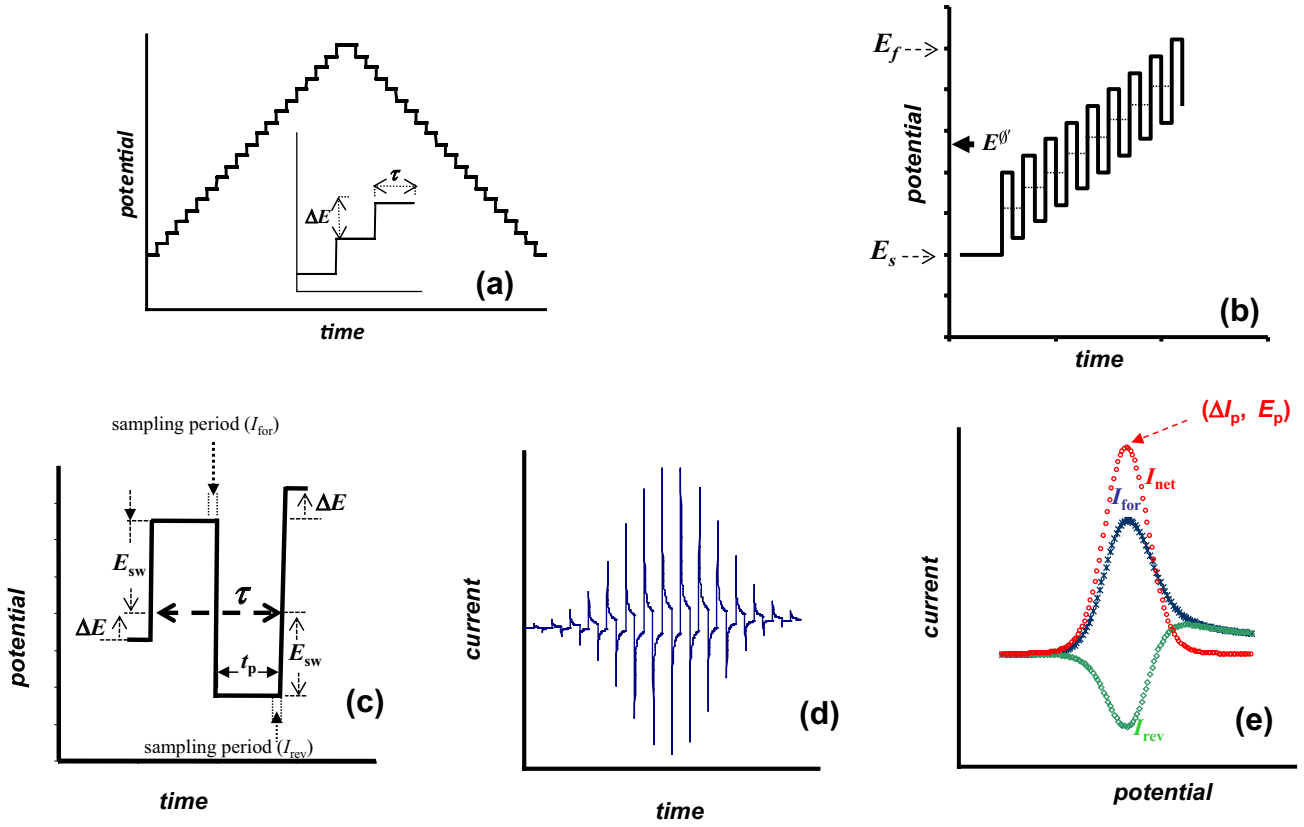


Fig. 5 **a** Potential wave-form in cyclic staircase voltammetry. The inset shows the duration of the potential step (τ) and the scan increment of the staircase potential (ΔE); **b** the potential modulation in square-wave voltammetry; **c** single potential cycle in square-wave voltammetry; **d** variation of the current in time in the course of the square-wave voltammetry; **e** typical square-wave voltammogram

micromolar bulk concentration). Thus, the charging current is the limiting factor in terms of the sensitivity of voltammetry in general [1].

Fortunately, for a given small potential step or potential increment (a few mV), the formation of the double layer typically is completed in a few milliseconds (or less) and the charging current (I_c) drops to negligible small values after such time interval, as depicted in the inset of Fig. 4. On the other hand, faradaic current (I_f) is still significant after certain time, as envisaged in the inset. Though both current components decay with time, the ratio I_f/I_c increases for a given potential, which is particularly important in the context of the following section.

Square-wave voltammetry

Stepwise potential variation is beneficial in voltammetry

From the foregoing discussion, one can plausibly assume that the sensitivity of voltammetry could be significantly improved by performing the experiment in the way to diminish the contribution of the charging current. This can be achieved by replacing the continuous potential ramp (cf. Fig. 3a) with a staircase potential-time function in cyclic voltammetry (Fig. 5a). Measuring the current at the end of each potential step diminishes the charging current contribution significantly, thus improving the sensitivity and quality of the voltammetric data profoundly. This sort of technique should be precisely termed as *current sampled cyclic staircase voltammetry* (CSCSV). The main features of the technique are the scan increment ΔE (cf. Fig. 5a) and the duration of each potential step τ. Obviously, the scan rate is defined as the ratio of the scan increment and the step duration, $v = \Delta E/\tau$.

Voltammetric experiment can be conducted even in more advanced fashion, if the staircase potential function is combined with small potential pulses, as depicted in Fig. 5b. The latter potential shape is typical for *square-wave voltammetry*, which is considered to be one of the most advanced forms of voltammetric techniques [5]. To understand the overall potential modulation, let us consider the potential variation during a single potential step (Fig. 5c). Indeed, at each potential step, two small, oppositely oriented potential pulses are imposed. The duration of the two pulses is

identical, designated as $t_p = \tau/2$, where τ, as previously defined, is the duration of the potential step. The height of a single pulse is termed as *square-wave amplitude* (E_{sw}). Thus, the main parameters of the SW potential modulation are the scan increment ΔE of the staircase ramp, the SW amplitude (E_{sw}) and the duration of the potential pulse t_p. Often, the later parameter is expressed in terms of the *square-wave frequency* $f = \frac{1}{2t_p}$. We can see latter, it is the critical time parameter of the SW voltammetric experiment. By analogy to cyclic staircase voltammetry, the SW potential wave-form can be attributed with scan rate defined as the product of the frequency and the scan increment, $v = f\,\Delta E$.

The two neighbouring potential pulses imposed at a single potential step complete a *potential cycle* in SWV (cf. Fig. 5c). The first and the second pulse are termed as *forward* and *reverse pulse*, respectively. The idea behind imposing two oppositely oriented potential pulses is to drive the electrode reaction (1) in both anodic and cathodic directions for the purpose of gaining mechanistic information; such procedure is repeated at each potential step, resulting in the overall potential modulation shown in Fig. 5b. One can inspect the mechanistic aspects of the electrode reactions at the potential value corresponding to each potential step, starting from the E_s to the final E_f potential, thus avoiding the necessity to reverse the overall potential modulation, as in CV.

The current variation in the course of the SW voltammetric experiment for the reaction (1) is depicted in Fig. 5d. During each potential cycle, one measures corresponding anodic and cathodic currents that reflect the shift of the electrode redox equilibrium in anodic and cathodic direction, respectively. For the sake of improved sensitivity the current is sampled at the end of each potential pulse only (cf. Fig. 5c). The series of current values measured at the end of each forward and reverse pulse constitute the forward (I_{for}) and reverse components (I_{rev}) of the SW voltammetric response, respectively (Fig. 5e). Obviously, at each potential step two current values are measured. Moreover, another voltammetric curve can be constructed by subtracting the corresponding values of the forward and reverse current values, termed as *net current,* $I_{net} = I_{for} - I_{rev}$.

Several justifications could be provided for such sort of simple mathematical transformation of the experimental data. For instance, in most of the cases, the net SW voltammogram is a well-defined peak, whereas the experimental forward and reverse components have a wave-like shape, or in some cases could have a more complicate shape. The peak-like shape of net SW component is advantageous in analytical context as it provides readily measurable parameters 'peak height' and 'peak position', thus improving the resolution of the technique. The intensity of the net response is higher compared to the forward and backward components, which is another important feature. Only in very specific cases, of completely irreversible electrode processes, the net SW peak can be smaller than the forward one (see Fig. 8, in the following "Quasireversible electrode reaction"). Finally, the contribution of the charging current to the net current may be even less significant than in particular forward and reverse components, for most of the cases. The reason is that the charging current measured at the end of the pulse is generally small. When small amplitudes are used ($nE_{sw} \leq 50$ mV; n—is the stoichiometric number of electrons), the small charging current at two neighbouring pulses is identical in the sign and vary close in the magnitude. Thus, the subtraction procedure in calculating I_{net} diminishes further the contribution of the charging current, making SWV one of the most sensitive voltammetric techniques [1, 5–7].

Typical SW voltammogram encompasses three *I–E* curves: two experimentally measured (forward and reverse), and the third calculated net current (Fig. 5e). Thus, to each potential of the staircase ramp, three current values are associated. Each voltammetric curve is discontinuous, and the point number depends on the scan increment. One should be aware that the forward and reverse currents are measured at the real potential of the corresponding pulses, $E_{for} = E_{step} + E_{sw}$, and $E_{rev} = E_{step} - E_{sw}$, for the forward and reverse pulses, respectively; they are, however, plotted versus the step potential E_{step}. Hence, the significant effect of the E_{sw} to the features of the response could be even intuitively predicted. The overall shape of the forward and reverse components resembles the cyclic voltammogram, and intrinsically they provide analogous mechanistic information as CV does. For all these reasons, one claims that SWV unifies the advantages of cyclic voltammetry, as a tool for mechanistic studies, with the advanced sensitivity, typical for pulse voltammetric techniques [8].

Reversible electrode reaction

One of the simplest electrode mechanisms in electrochemistry is the so-called *reversible electrode reaction* of a solution resident redox species, as described with Eq. (1) in "The simple shape of the voltammogram reflects a variety of interfacial phenomena". Besides all previous suppositions, let us further simplify the system by assuming that the stoichiometric number of electrons is $n = 1$, thus a one-electron reversible electrode reaction is considered.

The term "reversible" in the context of general chemistry most frequently means a two-directional chemical reaction system ($A \rightleftarrows B$). The reaction system $A \rightleftarrows B$ can reach the state of chemical equilibrium, which is dynamic state, where the rate of the forward ($A \rightarrow B$) and reverse ($B \rightarrow A$) chemical reactions are equal. In the thermodynamics, "reversible" refers to any process which goes from an initial to a final

state at an infinitesimally slow rate (thus requiring an infinite time), being considered to be permanently in the state of equilibrium. In voltammetry, reversible electrode reaction is the one in which the redox species at the interface reach the equilibrium concentrations at any potential. In other words, at the interface, the Nernst equation holds for any potential of the working electrode. The latter equation, written in terms of surface concentrations, is as follows:

$$E(t) = E^{\emptyset'} + \frac{RT}{F} \ln \frac{c_{\mathrm{Ox}}(0,t)}{c_{\mathrm{Red}}(0,t)}, \tag{3}$$

where $E^{\emptyset'}$ is the formal potential related to the standard potential of the redox couple and the activity coefficients of redox species. Alternatively, Nernst equation can be written as

$$c_{\mathrm{Ox}}(0,t) = c_{\mathrm{Red}}(0,t) \exp(\varphi), \tag{4}$$

where

$$\varphi = \frac{F}{RT}\left(E - E^{\emptyset'}\right), \tag{5}$$

where φ is the dimensionless potential, which can be understood as a driving force of the interfacial electron transfer process. As mentioned in "The simple shape of the voltammogram reflects a variety of interfacial phenomena", the notation $c(0,t)$ means the surface concentration. From the Nernst equation, we learn that the variation of the electrode potential $E(t)$, as an independent, instrumentally controllable variable, causes variation of the concentration ratio $\frac{c_{\mathrm{Ox}}(0,t)}{c_{\mathrm{Red}}(0,t)}$. It means a shift of the redox equilibrium (1) at the electrode surface manifested as a flow of electric current.

The Nernst equation (3) holds only at the electrodelelectrolyte interface and the overall system in the voltammetric cell is not at equilibrium. As explained in "The simple shape of the voltammogram reflects a variety of interfacial phenomena", the electrode reaction (1) makes the surface concentrations different than in the bulk, thus prompting spontaneous mass transfer by diffusion in the vicinity of the electrode. Consequently, the reversible electrode reaction in voltammetry is affected by both potential of the working electrode and the diffusion of redox species. In the interplay between the electrode potential and the mass transfer, the latter is the limiting step.

To establish a theoretical background, besides Nernst Eq. (3), Fick's law of diffusion has to be considered [3]. Unfortunately, diffusion is described by a complex mathematics, which is frequently beyond the common education in chemistry. Nevertheless, we are able to describe the experiment qualitatively in simple mathematical terms as follows. At the beginning of the experiment, $t = 0$, only Red is present in the system, being equally distributed in the electrolyte solution ($x \geq 0$) at a concentration $c_{\mathrm{Red}}(x,0) = c_{\mathrm{b,\,Red}}$, while $c_{\mathrm{Ox}}(x,0) = 0$.[2] When the experiments start ($t > 0$), the electrode reaction (1) takes place at the initial potential E_{s} (cf. Fig. 5b), giving rise to surface concentrations $c_{\mathrm{Red}}(0,t)$ and $c_{\mathrm{Ox}}(0,t)$ interconnected through the Nernst Eq. (3). As Red is consumed ($c_{\mathrm{Red}}(0,t) < c_{\mathrm{b,\,Red}}$ while Ox is formed ($c_{\mathrm{Ox}}(0,t) > c_{\mathrm{b,\,Ox}}$), Red diffuses from the bulk toward the electrode surface, while Ox diffuses in the opposite direction. The rate of diffusion depends on the concentration gradients established in the vicinity of the electrode and the diffusion rate constant, i.e., the diffusion coefficient D. The latter is commonly expressed in units of $\mathrm{cm^2\ s^{-1}}$, and the typical value is in order of $10^{-6}\ \mathrm{cm^2\ s^{-1}}$, for most of species in an aqueous electrolyte. Fortunately, the assumption that both redox species have the same diffusion coefficient is frequently close to the reality.

After complex mathematical modelling and computer simulations, the response of the one-electron reversible electrode reaction under conditions of SWV can be expressed as

$$I = A\Psi(\varphi), \tag{6}$$

where A could be termed as an *amperometric constant* defined as

$$A = FSc_{\mathrm{b,\,Red}}\sqrt{Df}, \tag{7}$$

where S is the electrode surface area, D is the common diffusion coefficient, and f is the frequency. $\Psi(\varphi)$ is the dimensionless function, which could be revealed through mathematical modelling and simulations only. It represents the effect of the potential wave-form and the diffusion of redox species in the course of the experiment. Thus, it can be intuitively assumed that $\Psi(\varphi)$ depends on the amplitude (E_{sw}) and scan increment (ΔE), as typical features of the potential modulation. Most importantly, $\Psi(\varphi)$ does not depend of the frequency and concentration of the redox species for a reversible electrode reaction. It means that the real current I, including the net peak-current ΔI_{p} (cf. Fig. 5e), is a *linear function of the square-root of the frequency*, which is the first diagnostic criterion of a reversible electrode reaction of a dissolved redox couple. In the reality, one varies the frequency typically from the minimal instrumental value [e.g., $f = 5$ Hz ($t_{\mathrm{p}} = 100$ ms) to 100 Hz ($t_{\mathrm{p}} = 5$ ms)]. Beyond the latter interval, deviations from linearity frequently occur for most of the experimental systems, the reason of which is discussed in next section.

Understanding the voltammetric behaviour of particular electrode mechanism commonly requires the study of the dimensionless function. The evolution of $\Psi(\varphi)$ for different SW amplitudes is depicted in Fig. 6. It consists of

[2] Strictly, there is always a certain concentration of Ox existing in a reversible equilibrium; however, in analytical context, its concentration is negligible to Red.

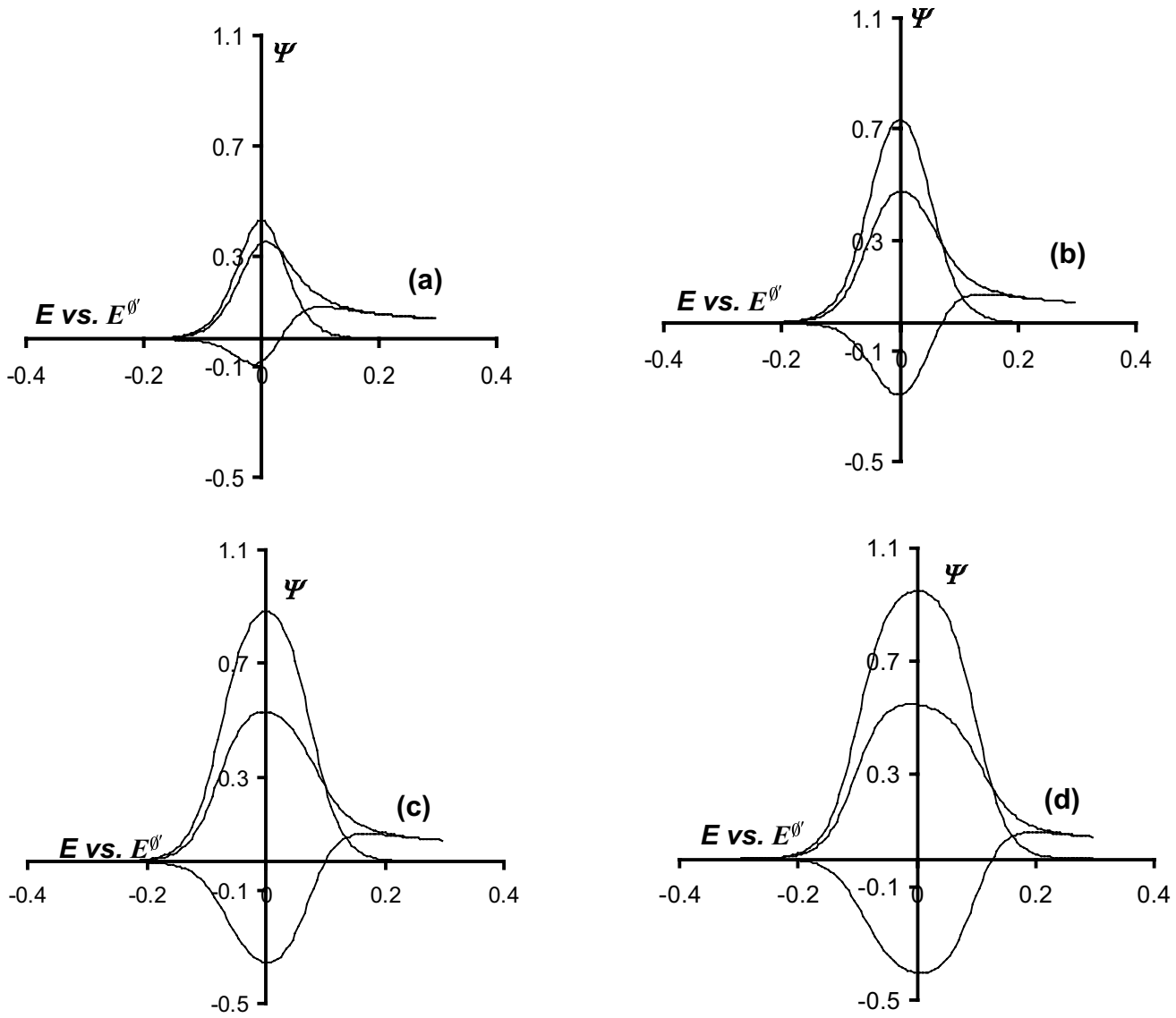


Fig. 6 Evolution of the dimensionless function $\Psi(\varphi)$ for a one-electron reversible electrode reaction calculated for the SW amplitude of 25 (**a**); 50 (**b**); 75 (**c**) and 100 mV (**d**). The scan increment is $\Delta E = 5$ mV

well-developed forward and reverse current components, and the net peak with the peak-potential identical to the formal potential of the electrode reaction (1). Taking into account Eq. (7) and recalling that $\Psi(\varphi)$ is independent of the frequency, the real net peak has the *peak-potential identical to the formal potential* of the electrode reaction, *independent of both SW amplitude and frequency*, which is the second diagnostic criterion of the reversible electrode reaction in SWV.

More detailed analysis of the morphological evolution of the response under the influence of the amplitude is relatively complex (cf. Fig. 6). For the typical amplitude of $E_{sw} = 50$ mV (Fig. 6b), the forward and reverse components have the same peak potentials, equal to the formal potential, while their peak-currents are different. The half-peak width of the net peak is 120 mV. For $E_{sw} < 50$ mV, the forward (anodic) component is shifted toward more positive potentials than the reverse (cathodic) one, while for $E_{sw} > 50$ mV, the interposition is reversed. At the same time, by enhancing the E_{sw} the intensity of the overall response increases, together with the half-peak width. However, similar effect on the morphological evolution of the voltammetric response is observed for many other mechanisms [5], implying that the influence of the amplitude cannot be used as a selective diagnostic criterion for characterising reversible electrode reaction.

Scan increment affects the scan rate of the overall waveform, thus the time of the experiment increases by decreasing the scan increment. Commonly, the scan increment is varied

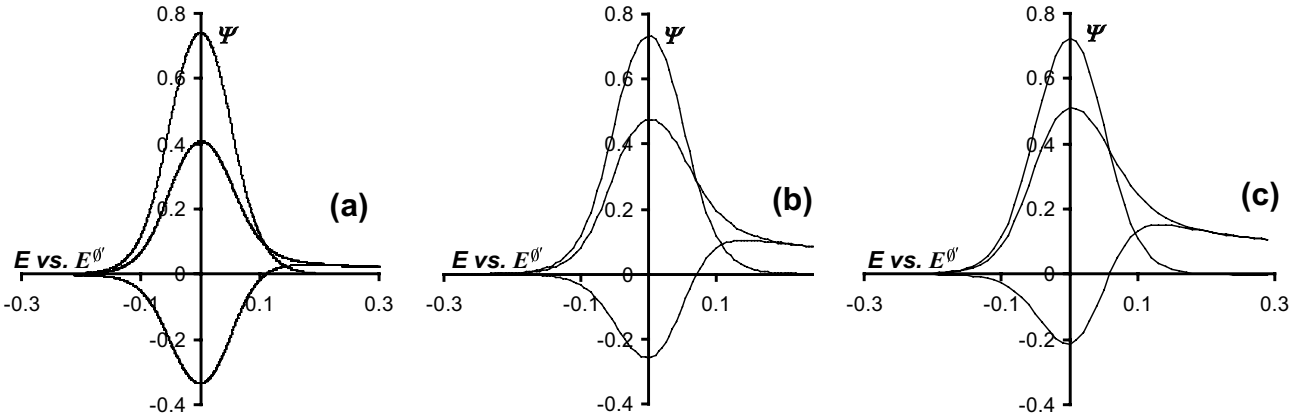


Fig. 7 Evolution of the dimensionless function $\Psi(\varphi)$ for a one-electron reversible electrode reaction calculated for the amplitude of $E_{sw} = 50$ mV and the scan increment of 0.5 (**a**); 5 (**b**), and 10 mV (**c**)

over the interval from 1 to 10 mV. The net peak-potential is independent on ΔE, while the net peak-current slightly decreases by increasing ΔE. Figure 7 depicts the morphological variation $\Psi(\varphi)$ for very low ($\Delta E = 0.5$ mV), moderate ($\Delta E = 5$ mV), and a large value ($\Delta E = 10$ mV) of the scan increment, revealing a marked influence to both forward and reverse components. For very low scan increment, when the overall time of electrolysis is long, the thickness of the diffusion layer significantly increases, making the rate of diffusion very slow at the end of the experiment. As a consequence both forward and reverse current components (Fig. 7a) gain a peak-like shape, contrary to the wave-like shape, associated to the diffusion tail for the large scan increment and the fast scan rate (Fig. 7c).

Quasireversible electrode reaction

In the experimental reality reversible electrode reactions are seldom encountered. Often, the electrode reaction is affected by the kinetics of the interfacial electron transfer; such electrode reaction is termed as *quasireversible*. Let us keep considering a one-electron electrode reaction (1), assuming that the oxidation (anodic; Red → Ox + e) and reduction (cathodic; Ox + e → Red) are attributed with anodic and cathodic electrochemical rate constant k_a and k_c, respectively. The electrochemical rate constants depend on the electrode potential. Hence, they are different than the rate constants of a common chemical reaction [3, 9]. The exact mathematical formulations are as follows:

$$k_a = k^{\emptyset'} \left[\exp(\varphi)\right]^{\beta}, \tag{8}$$

$$k_c = k^{\emptyset'} \left[\exp(-\varphi)\right]^{\alpha}, \tag{9}$$

Here, φ is the dimensionless electrode potential, i.e., the driving force of the interfacial electron transfer, being identical as in the context of the reversible electrode reaction (Eqs. 3–5). In the case of the reversible electrode process, the driving force φ dictates instantaneous redox equilibrium at the electrode surface. For the quasireversible reaction, the same driving force determines the rate to reach the equilibrium, which is not an instant process, but requires certain time. Of course, if the rate is low and the time is short, the redox equilibrium is not established, which is typical situation for the quasireversible electrode reaction.

In the particular case when the electrode potential is $E = E^{\emptyset'}$, the driving force is $\varphi = 0$; thus $\exp(0) = 1$, and both rate constants are equal, $k_a = k_c = k^{\emptyset'}$. The latter parameter, $k^{\emptyset'}$, known as the *formal rate constant*, is an intrinsic kinetic parameter of electrode reactions [9]; $k^{\emptyset'}$ has dimensions of length per time, due to the heterogeneous nature of the electron transfer occurring across the interface.

Several explanations can be offered to understand the meaning of the formal rate constant. For instance, at a given driving force φ, an electrode reaction attributed with small $k^{\emptyset'}$ requires longer time to reach the equilibrium compared to a reaction attributed with larger $k^{\emptyset'}$. Alternatively, one can say that an electrode reaction with small $k^{\emptyset'}$ requires larger driving force φ than a reaction with larger $k^{\emptyset'}$ to gain an identical rate. A useful concept is to consider an electrode reaction in equilibrium at the formal potential $E = E^{\emptyset'}$. From the Nernst Eq. (3) the surface concentrations of redox species are equal, $c_{Ox}(0,t) = c_{Red}(0,t) = c(0,t)$. However, the equilibrium is dynamic process, and both anodic and cathodic reactions must proceed at equal rate, which fortunately is true, considering Eqs. (8) and (9), i.e.,

$$v_a = v_c = v = k^{\emptyset'} c(0,t). \tag{10}$$

Yet, there is a significant discrepancy between two reactions attributed with different formal rate constants in the state of equilibrium. From Eq. (10), an electrode reaction with small $k^{\varnothing'}$ has a small exchange rate between Ox and Red under equilibrium conditions, compared to another electrode reaction attributed with larger $k^{\varnothing'}$, where the turnover between Ox and Red is faster.

In Eqs. (8) and (9), exponents α and β are dimensionless numbers related as $\alpha+\beta=1$, known as cathodic and anodic electron transfer coefficients, respectively [4, 9]. For the sake of simplicity, it is sufficient to define only a single electron transfer coefficient, e.g., the anodic one β and the cathodic coefficient is $\alpha=1-\beta$. Thus, the rate constants can be written as $k_a = k^{\varnothing'}\left[\exp(\varphi)\right]^{\beta}$ and $k_c = k^{\varnothing'}\left[\exp(-\varphi)\right]^{(1-\beta)}$. Commonly, for most of the electrode processes $\alpha=\beta=0.5$. If $\beta \neq 0.5$, then an asymmetry appears in the degree of variation of the rate constants k_a and k_c for the same driving force (in absolute value) φ of the electrode reaction. For these reasons, β is frequently termed as a symmetry factor in the context of electrode kinetics [9].

In modelling of a quasireversible electrode reaction under voltammetric conditions, the most important equation is the Butler–Volmer equation, which relates the current to $k^{\varnothing'}$, the driving force φ, and the surface concentrations of redox species:

$$\frac{I}{FS} = k^{\varnothing'}\left[\exp(\varphi)\right]^{\beta}\left[c_{\mathrm{Red}}(0,t) - c_{\mathrm{ox}}(0,t)\exp(-\varphi)\right]. \quad (11)$$

It is derived taking into account Eqs. (8) and (9), and considering that the net-rate of a heterogeneous electrode reaction at a given driving force ($v = v_a - v_c$) is equivalent to the ratio $\frac{I}{FS}$. The latter is known as flux (in units of mol s^{-1} cm^{-2}), i.e., the amount of a reacted material per unit of time and surface area. In addition, the mass transfer by diffusion has to be again taken into account, equivalent as for the reversible electrode reaction. In the case of a quasireversible electrode reaction, however, the rate of the electron transfer is the limiting step in the interplay between the electron and mass transfer for the major portion of the voltammetric experiment.

The voltammetric response under conditions of SWV can be expressed as

$$I = A\Psi(k^{\varnothing'}, \varphi, f, D), \quad (12)$$

where A is the amperometric constant defined as for the reversible electrode reaction (Eq. 7). Yet, the dimensionless function is rather complex, depending on $k^{\varnothing'}, \varphi$, the frequency f (i.e., the duration of potential pulses), and the rate constant of the diffusion D (i.e., the diffusion coefficient). Fortunately, in the exact mathematical formula for the dimensionless function Ψ, the formal rate constant $k^{\varnothing'}$, frequency f, and the diffusion coefficient D are unified in a single *dimensional electrode kinetic parameter* $\kappa = \frac{k^{\varnothing'}}{\sqrt{Df}}$, which is of critical importance for understanding voltammetric behaviour [5].

First of all, let us note that κ unifies the effect of the electron exchange rate ($k^{\varnothing'}$), the rate of mass transfer (D), and the critical time of the voltammetric experiment (f). Figure 8 shows sigmoid variation of the net peak-current with the logarithm of the dimensionless kinetic parameter κ. At first glance, Fig. 8 should be understood as a comparison of a series of electrode reactions attributed with different formal rate constants $k^{\varnothing'}$, while other experimental conditions, including D and f, are identical. Roughly, in electrochemical terms, three kinetic regions can be identified, i.e., *irreversible* [$\log(\kappa) < -1.5$], *quasireversible* [$-1.5 \leq \log(\kappa) \leq 1$] and *reversible* [$\log(\kappa) > 1$]. In the reversible kinetic region, $\log(\kappa) > 1$, the electrode kinetics is fast enough to enable the redox equilibrium to be established in the course of the potential pulse. Accordingly, the diffusion is the rate limiting step and the electrode reaction has an electrochemically reversible behaviour. The dimensionless function $\Psi(k^{\varnothing'}, \varphi, f, D)$ (Eq. 12) is becoming identical with the function $\Psi(\varphi)$ for the reversible reaction (Eq. 6). The inequality $\log(\kappa) > 1$ can be rewritten in the following form:

$$k^{\varnothing'} > 10\sqrt{Df}, \quad (13)$$

which has a very useful physical meaning. The left-hand side $\left(k^{\varnothing'}\right)$ represents the interfacial electron transfer kinetics, while the right-hand side $\left(\sqrt{Df}\right)$ represents the mass transfer kinetics (i.e., diffusion). Equation (13) is an important criterion showing in what relation should be the kinetics of the electron transfer versus diffusion rate to exhibit a reversible electrochemical behaviour.

Accordingly, the criterion for a quasireversible behaviour can be written as $0.032\sqrt{Df} \leqslant k^{\varnothing'} \leqslant 10\sqrt{Df}$. Within this region, the current is determined by the electrode kinetics and the interfacial electron transfer is slower than the mass transfer. Considering a single voltammogram, the latter statement is valid for the first part of the net SW voltammogram, including the net peak-current. At potentials beyond the net peak-current, the driving force φ of the forward reaction ($\mathrm{Red} \rightarrow \mathrm{Ox} + e^-$) is becoming very large, making the kinetics of electron transfer again faster than the diffusion, which is the reason for expansion of the diffusion layer and exhaustion of the initial electroactive material Red. Whereas, for $k^{\varnothing'} < 0.032\sqrt{Df}$ the electrode kinetics is sluggish, causing the amount of formed Ox species at the electrode surface to be minute and the rate constant of the reverse reaction $\left(k_c = k^{\varnothing'}\left[\exp(-\varphi)\right]^{(1-\beta)}\right)$ to be negligible for any φ, due to very small value of $k^{\varnothing'}$. As a consequence, the reverse (cathodic; $\mathrm{Ox} + e \rightarrow \mathrm{Red}$) reaction does not proceed at a measurable rate at any potential;

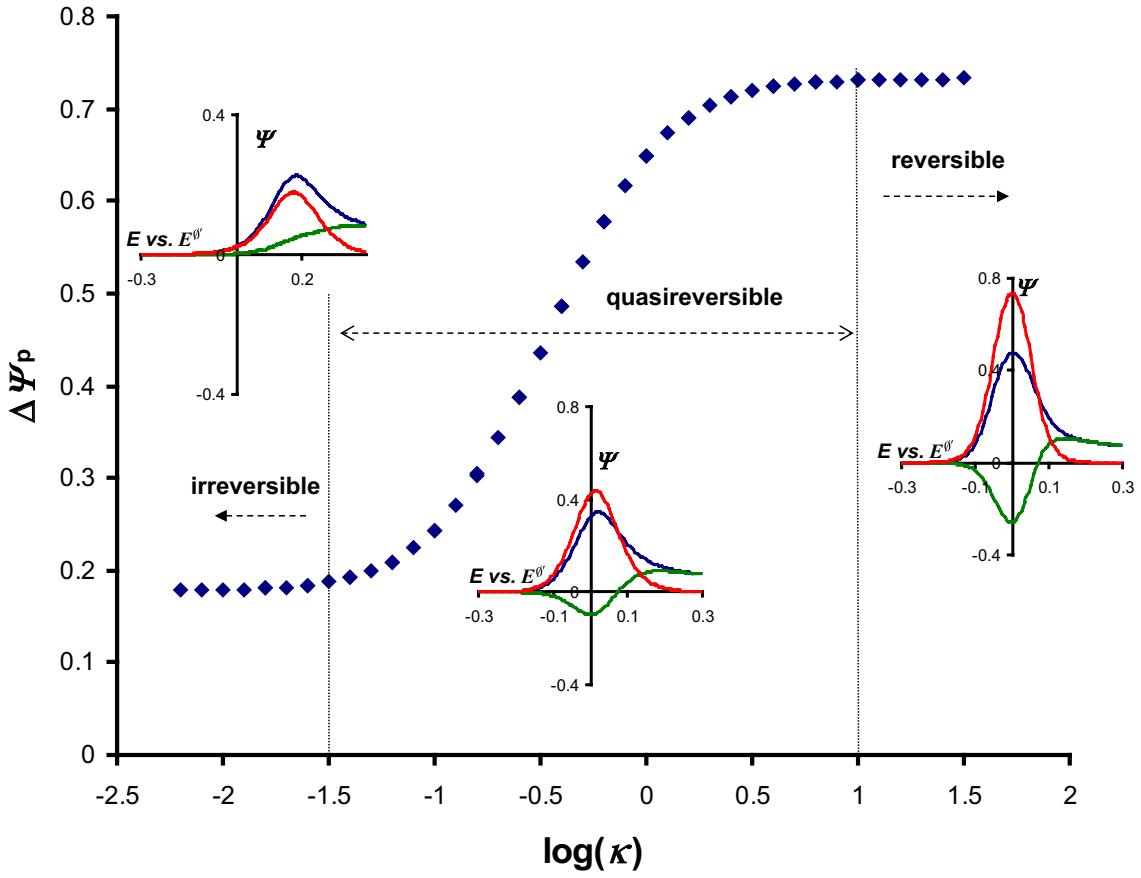


Fig. 8 Variation of the net peak-current of the dimensionless function Ψ of a quasireversible electrode reaction with the electrode kinetic parameter κ, for amplitude $E_{sw}=50$ mV and scan increment $\Delta E=5$ mV. The inset shows the dimensionless SW voltammetric response typical for the irreversible (left), quasireversible (middle) and reversible (right) kinetic region

hence the reverse component of the response is missing (see the inset at the left-hand side of Fig. 8). For this reason, such electrode reaction is termed as *electrochemically irreversible*.

Besides remarkable variation of the net peak-current and the morphology of the overall response with κ (cf. Fig. 8), the position of the response at the potential axis is strongly sensitive to the electrode kinetics. Generally, the response is shifted toward more positive potentials versus the formal potential $E^{\emptyset\prime}$ by decreasing the electrode kinetics, for the case of the oxidative electrode mechanisms (Eq. 1), in which the experiments start in the presence of Red species only. Accordingly, the shift of the response will be in the opposite direction (toward more negative values relative to $E^{\emptyset\prime}$) for reductive electrode mechanism ($Ox+e^{-}\rightleftarrows Red$). Hence, the variation of the net peak-potential with the electrode kinetics is an *important criterion for characterisation of a quasireversible electrode reaction*.

Electrode kinetic parameter $\left(\kappa=\frac{k^{\emptyset\prime}}{\sqrt{Df}}\right)$ unifies two typical constants of a given experimental system ($k^{\emptyset\prime}$ and D) with the frequency of the potential modulation. Hence, the degree of reversibility of a single electrode reaction can be varied by the frequency (i.e., duration of potential pulses). For illustration, let us assume an electrode reaction with $k^{\emptyset\prime}=0.08$ cm s^{-1} and $D=5\times10^{-6}$ cm^2 s^{-1}. At $f=5$ Hz, $\log(\kappa)=1.204$; thus the reaction behaves as a reversible process (cf. Fig. 8). In other words, the duration of a single potential pulse is long enough to achieve the redox equilibrium at the end of the pulse [i.e., when the current is sampled (cf. Fig. 5c)], whereas, for $f=1000$ Hz, $\log(\kappa)=0.054$ and the reaction behaves as a typical quasireversible process, in which the kinetics of the electron transfer is the rate determining step. Obviously, the variation of the frequency from 5 to 1000 Hz transposes the system from the reversible into the quasireversible kinetic region.

The frequency exhibits even more complex effect to the real current (Eq. 12), as it affects simultaneously both $\Psi(k^{\emptyset\prime},\varphi,f,D)$ and the amperometric constant $\left(A=FSc_{b,\mathrm{Red}}\sqrt{Df}\right)$. For these reasons, the real net peak-current is complex, non-linear function on the square-root

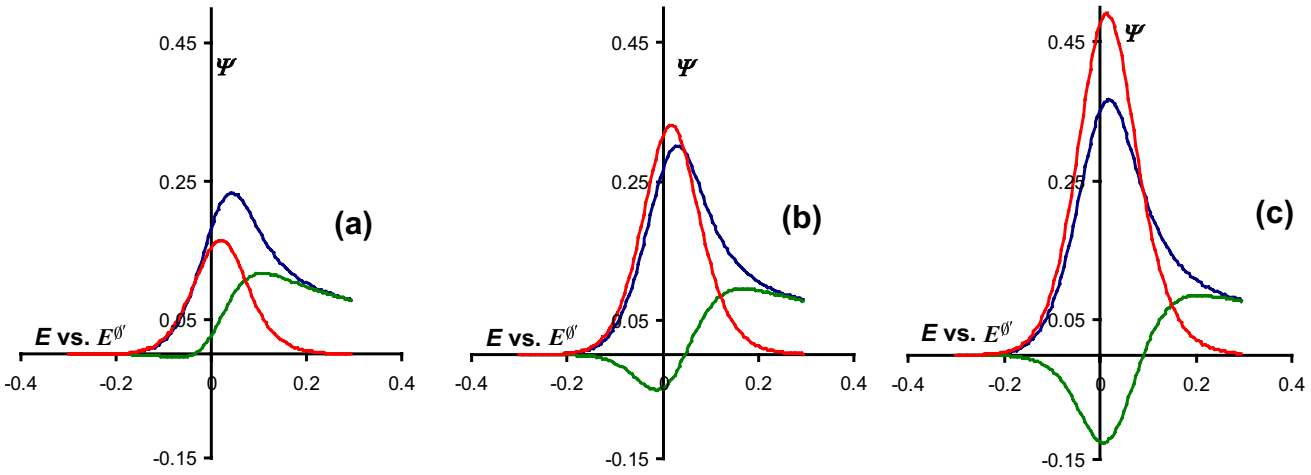


Fig. 9 Evolution of the dimensionless function Ψ of a quasireversible electrode reaction characterised with $k^{0\prime}=2\times10^{-3}$ cm s^{-1}, $\beta=0.5$, and $D=5\times10^{-6}$ cm^2 s^{-1} with the amplitude of $E_{sw}=20$ (**a**); 40 (**b**) and 60 (**c**) mV. The frequency and the scan increment are $f=10$ Hz and $\Delta E=5$ mV

of the frequency, contrary to the reversible electrode reaction. So, the variation of the morphology of the response, and the net peak-current and potential with the frequency is the clear criterion for differentiation of the quasireversible and reversible electrode reactions.

Unfortunately, the voltammetric behaviour of the quasireversible electrode reaction is even more complex and the electrochemical reversibility is not solely determined by the electrode kinetic parameter κ. The above given criterion for the quasireversible kinetic region $\left(0.032\sqrt{Df} \leqslant k^{0\prime} \leqslant 10\sqrt{Df}\right)$ is actually valid for the conditions of Fig. 8, i.e., the amplitude $E_{sw}=50$ mV and the scan increment $\Delta E=5$ mV. Changing these intrinsic parameters of the SW potential wave-form affects the quasireversible kinetic region and the actual interrelation between $k^{0\prime}$, D and f. Let us recall that the amplitude determines the exact value of the pulse potential (cf. Fig. 5c). Thus, E_{sw} affects the driving force of the electrode reaction φ and finally, in a combination with $k^{0\prime}$ and the surface concentrations $c(0,t)$ (Eq. 11), determines the actual rate of the electrode reaction [10]. Figure 9 clearly illustrates how the increasing amplitude transposes a single electrode reaction from the irreversible to the quasireversible kinetic region, for a given experimental conditions (i.e., constant frequency and scan increment).

Finally, a similar effect is expected to be observed under the influence of the scan increment for a single electrode reaction at a given set of values for E_{sw} and f. As mentioned in "Stepwise potential variation is beneficial in voltammetry", the overall scan rate of the SW potential wave-form is defined as $v=f\Delta E$. The scan rate determines the overall time of the experiment, the expansion of the diffusion layer, and the rate of the diffusion. Thus, it is expected to exhibit an influence on the interrelation between $k^{0\prime}$ and $\sqrt{Df}$ in terms of the electrode kinetic regions. The influence of ΔE on the electrode kinetic is yet insufficiently elaborated in the scientific literature, which remains to be a challenge for further studies.

Conclusions

Square-wave voltammetry is certainly one of the most advanced and versatile members in the family of pulse voltammetric techniques. It possesses high analytical sensitivity and speed of measurements. The excitation signal, i.e., potential modulation in the course of the voltammetric measurement, enables the electrode reaction to be driven in both oxidative and reductive directions repeatable at each step of the staircase potential, thus providing an insight into the mechanistic aspects of the studied electrode reaction. Hence, square-wave voltammetry combines the advantageous of pulse techniques, with respect to the high analytical sensitivity, and cyclic voltammetry, with respect to the electrode mechanism insight. With respect to the analytical performances it is quite comparable to differential pulse voltammetry. However, SWV is yet superior in regard to electrode kinetics measurements and mechanistic aspects of electrode processes.

The technique is generally fast. Typical time intervals for driving the electrode reaction in both cathodic and anodic directions are in millisecond range. Hence, it is more suited to analyse reversible or quasireversible electrode processes and electrode processes coupled with fast chemical reactions, than to study sluggish electrode processes. When the

electrode reaction is very slow, square-wave voltammetry provides even smaller response than other simpler form of voltammetry, e.g., linear sweep voltammetry, which is the main disadvantage of the technique. Therefore, cyclic voltammetry is yet unavoidable in the preliminary mechanistic study of particular electrode mechanism. The strong synergy between cyclic and square-wave voltammetry is achieved in coupling both techniques in a form known as cyclic square-wave voltammetry. Finally, it is worth mentioning that square-wave voltammetry is particularly powerful technique when electrode processes are coupled with adsorption phenomena, which could be the subject of elaboration of forthcoming chem-texts.

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